

**AMENDMENT NO. 2 MARCH 2005
TO
IS 4859 : 1968 SPECIFICATION FOR
HIGH STRENGTH AIRCRAFT PLYWOOD**

(Page 7, clause 3.2, third line) — Substitute 'IS 848 : 1974*' for 'IS : 848 - 1957*'.

(Page 7, footnote marked *) — Substitute the following for the existing footnote:

'*Specification for synthetic resin adhesives for plywood (phenolic and aminoplastic) (first revision)'

(Page 11, clause 5, and page 12, Table 2) — Substitute the following for the existing:

5 DIMENSIONS AND TOLERANCES

5.1 The dimensions of plywood panels shall be as follows:

1 800 mm × 1 200 mm	1 800 mm × 900 mm
1 500 mm × 1 200 mm	1 500 mm × 900 mm
1 200 mm × 1 200 mm	1 200 mm × 900 mm

5.2 Thickness

The thickness shall be 1.0 mm, 1.5 mm, 2.0 mm, 2.5 mm, 3.0 mm, 3.5 mm, 4.0 mm, 4.5 mm, 5.0 mm, 6.0 mm, 7.0 mm, 8.0 mm, 9.0 mm and 10.0 mm.

NOTE — Any other dimensions (length, width and thickness) as agreed to between the manufacturer and the purchaser may also be used.

5.3 Tolerances

Tolerances on the nominal sizes of finished boards shall be as given below:

<i>Dimension</i>	<i>Tolerance</i>
Length	+6 -0 mm
Width	+3 -0 mm

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Thickness

- | | |
|--------------------|------------------|
| i) Less than 6 mm | ± 10 percent |
| ii) 6 mm and above | ± 5 percent |

Edge straightness

2 mm per 1 000 mm or 0.2 percent

Squareness

2 mm per 1 000 mm or 0.2 percent

NOTE — Edge straightness and squareness shall be tested as per Annex A

(Page 12, clause 7.1) — Substitute 'IS 1734 (Part 5) 1983*' for 'Appendix A of IS 709 1957*' in the fifth line and 'Annex B' for 'Appendix A' in the sixth line

(Page 12, footnote) — Substitute the following for existing footnote

*Method of test for plywood Part 5 Test for adhesion of plies

(Page 13, clause 9.1) — Substitute 'Annex C' for 'Appendix B' in second line.

(Page 13, clause 10.1) — Substitute 'Annex D' for 'Appendix C' in first line

(Page 13, clause 10.2.2) — Substitute 'Annex E' for 'Appendix D' in second line

(Page 13, clause 10.2.3) — Add the following new clause after 10.2.3.

10.3 Static Bending Strength — Three test specimens taken in each direction from the sample of plywood, when tested in accordance with IS 1734 (Part 11) . 1983 shall have an average and a minimum individual Modulus of Elasticity and Modulus of Rupture not less than the values specified in Table 2

Table 2 Modulus of Elasticity (MOE) and Modulus of Rupture (MOR)

Direction	MOE (N/mm ²)		MOR (N/mm ²)	
	Average	Min Ind	Average	Min Ind
(1)	(2)	(3)	(4)	(5)
Along (Direction parallel to the grain direction of the face veneer)	7 500	6 700	50	45
Across (Direction perpendicular to the grain direction of the face veneer)	4 000	3 600	30	27

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(*Page 14, clause 13.2.2*) — Insert the following Annex A after clause 13.2

ANNEX A
(*Clause 5.3*)

**METHOD OF TEST FOR EDGE STRAIGHTNESS AND
SQUARENESS**

A-1 PROCEDURE FOR EDGE STRAIGHTNESS

A-1.1 The straightness of the edges and ends of plywood shall be verified against a straight edge not less than the full length of the plywood. If the edge on the end of the plywood is convex, it shall be held against the straight edge in such a way as to give approximately equal gap at each end. The largest gap between the straight edge and the edge shall be measured to the nearest millimetre and recorded.

A-2 PROCEDURE FOR SQUARENESS

A-2.1 The squareness of plywood shall be checked with a 1 200 mm × 1 200 mm square, by applying one arm of the square to the plywood. The maximum width of the gap shall be recorded.

(*Page 15, Appendix A*) — Rename '**Appendix A**' as '**Annex B**'

(*Page 16, Appendix B*) — Rename '**Appendix B**' as '**Annex C**'

(*Page 17, Appendix C*) — Rename '**Appendix C**' as '**Annex D**'

(*Page 18, Appendix D*) — Rename '**Appendix D**' as '**Annex E**'

(CED 20)

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